Summer Training / Internship Project Report

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(Stock Span Analyzer using Stack)

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Introduction

1.Objective

1. **To understand and implement** the stock span problem using stack data structures.
2. **To develop** an efficient algorithm that calculates the stock span for a series of daily stock prices.
3. **To analyze** the performance of the stack-based approach in terms of time and space complexity.

4. **To demonstrate** how stacks can be applied to solve real-world financial problems like stock analysis.
5. **To provide** a user-friendly interface (if applicable) for users to input data and view results effectively.

Training Overview

1.Tools and Technologies used

Programming Language:

- **C++** – for implementing the stock span algorithm using stack

Data Structures:

- **Stack** – for efficient calculation of stock span values

Development Environment:

- **VS Code** /Dev++

Version Control:

- **Git** and **GitHub** – for source code management and collaboration

Project Details

1.Title of the Project

Stock Span Analyzer Using Stack

2.Problem Definition

In financial markets, the stock span problem is a common algorithmic challenge: for each day in a series of daily stock prices, determine the number of consecutive previous days the stock price was less than or equal to the current day's price.

A brute-force approach may result in $O(n^2)$ time complexity. The goal is to build an efficient algorithm using a stack to reduce the complexity to $O(n)$, making it scalable for large datasets.

3. Scope and Objectives

This project focuses on implementing a stack-based solution to analyze stock spans. It demonstrates how a linear-time algorithm can improve performance in stock analysis and provides a basic interface or output format for user interaction.

Objectives:

- To understand and implement the stock span algorithm using stack.
- To optimize performance for analyzing large sets of stock prices.
- To design a simple interface or console output for displaying results.
- To test and validate the solution with various datasets.

4. System Requirements

Software Requirements:

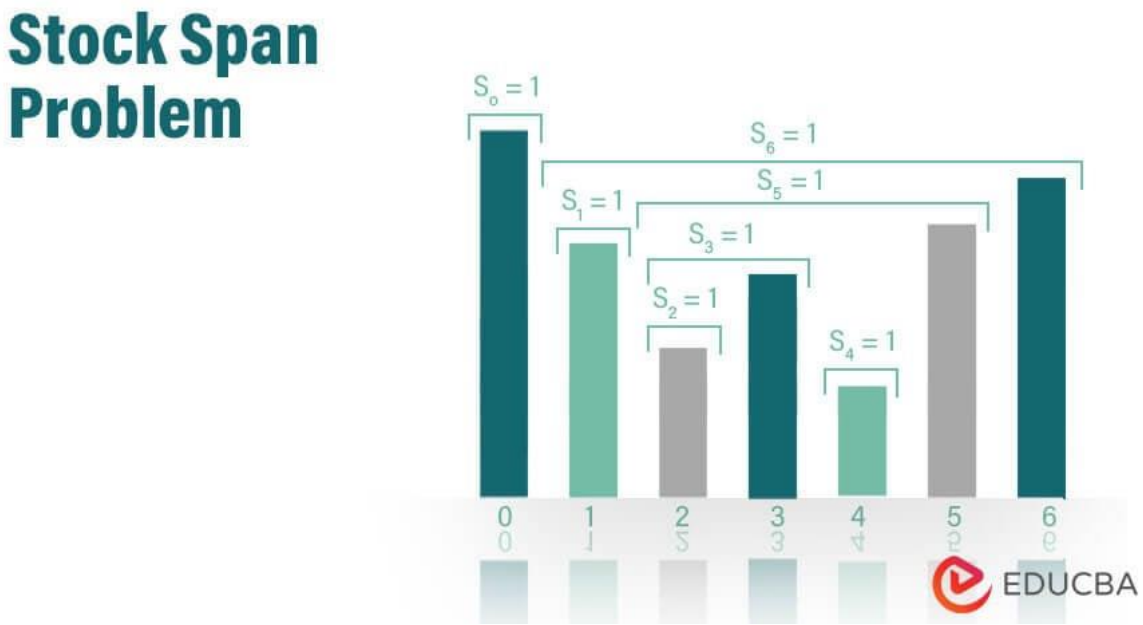
- Programming Language: C++ (as applicable)
- IDE: VS Code / PyCharm / Eclipse

- OS: Windows / Linux / macOS

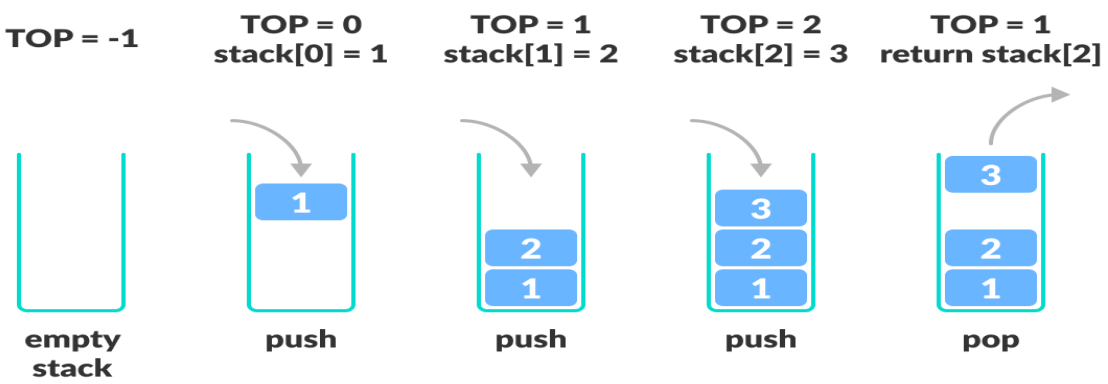
Hardware Requirements:

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Hard Disk: 100 MB free space

5. Architecture Diagram



6.Data flow diagram

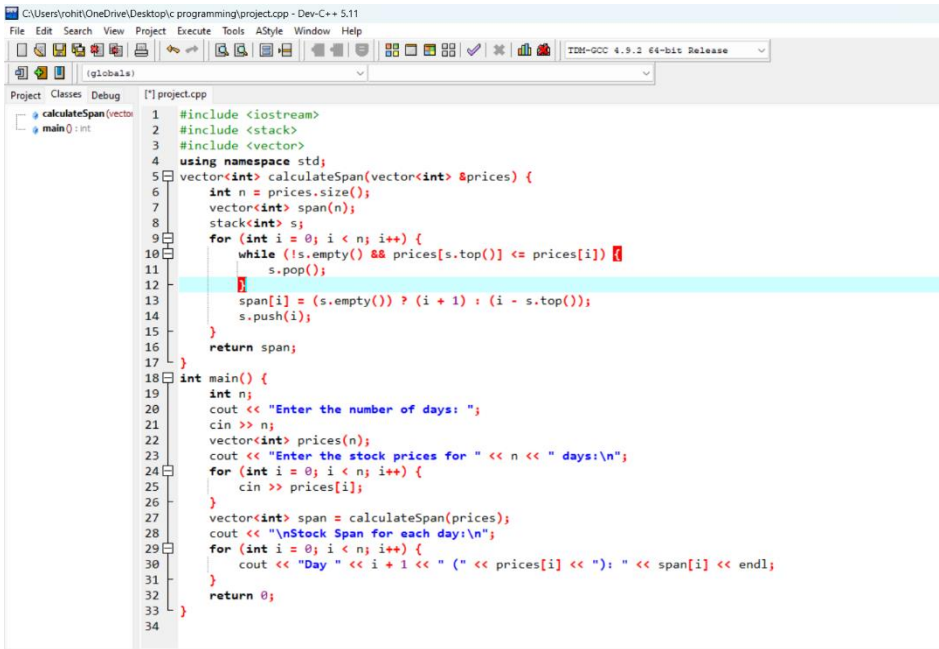


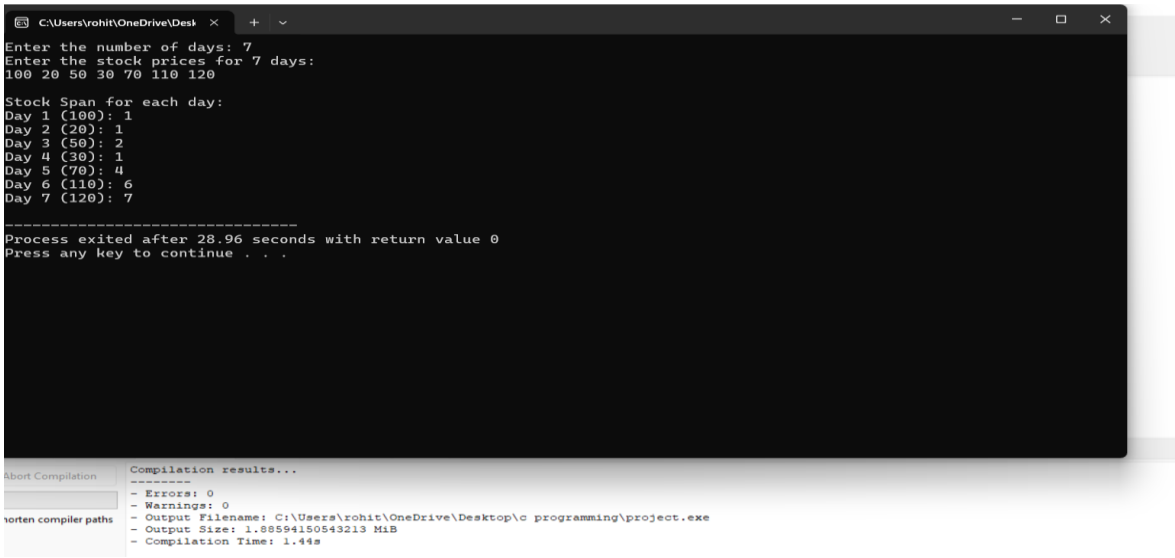
Implementation

1.Methodology

- Step 1: Input the array of stock prices.
- Step 2: Initialize a stack to keep track of indices of the stock prices.
- Step 3: Traverse the price array and calculate the span using the stack:
    - While the stack is not empty and the current price is higher than the price at the index stored in the stack, pop elements.
    - The span is the difference between the current index and the index on the top of the stack.
    - Push the current index to the stack.
- Step 4: Display span for each day.

2.Screenshot





Results and Discussion

1.Output

- Input: [100, 20, 50, 30, 70, 110, 120]
- Output Span: [1, 1, 2, 1, 4, 6, 7]
- Execution Time: Very efficient (O(n) time)

2.Challenges Faced

- Understanding and implementing stack logic efficiently
- Handling edge cases (e.g., all increasing or decreasing prices)
- Creating dynamic UI for visual output (if GUI involved)
- Debugging index errors in logic

3.Learnings

- Strengthened understanding of stack data structure
- Hands-on experience with real-world stock analysis use-case
- Improved debugging and algorithmic thinking
- Integration of logic with frontend (if applicable)

Conclusion

1.Summary

- The Stock Span Analyzer was successfully implemented using stack data structure.
- The program efficiently computes the span of stock prices in linear time O(n).
- It demonstrates a classic use-case of stacks in algorithmic problem-solving.
- The tool is useful for understanding how past stock performance affects current pricing trends.